

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Contello.ai | *TypeScript/Node.js, AWS, Playwright, OpenAI*

Jan. – Dec. 2025

Software Engineer Intern

Toronto, ON, Canada

- Integrated **OpenAI** into production **TypeScript/Node.js** on **AWS**; built **agentic content-generation workflows** serving **thousands/day** with **observability** via **CloudWatch**.
- Enabled **AI-assisted content creation** by ingesting brand docs via an **NLP** pipeline, caching **embeddings**, and auto-injecting context while enforcing **responsible AI guardrails**.
- Strengthened **AI reliability** to **99.9%** via **OpenAI Responses API** migration, **cost/latency tracking**, **retry/backoff**, and **Playwright E2E** gates in **CI/CD**.

RBC Royal Bank | *Apache Spark, Python, Airflow, SQL, Linux*

Jan. 2026 – Present

Software Developer Co-op

Vancouver, BC, Canada

- Engineered **Python/Airflow** pipelines processing **50M+ daily transactions** from **Delta Lake**, enabling downstream **ML/AI teams** with analysis-ready data on **petabyte-scale** financial systems.
- Optimized **Spark** jobs by migrating to **Spark-native APIs** and tuning **AQE/cluster memory**, cutting **latency 40%** and **compute costs** via **resource profiling**.
- Developing **multi-region** pipelines with **observability**, **fault tolerance**, and automated **failover**; instrumenting **cost/performance metrics** to inform infra decisions.

RESEARCH EXPERIENCE

UBC Emerging Media Lab | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

Software Developer & Team Lead

Vancouver, BC, Canada

- Led a **human-AI collaboration** research project: built an **AI-driven VR** simulation in **Unreal (C++)** pairing nurse practitioners with AI patients for clinical training.
- Designed a **multithreaded supervisory architecture (C++/Python)** enabling **real-time AI monitoring** and protocol enforcement across **agentic** patient actors.
- Analyzed **AI system performance**, achieving **40%** latency reduction and **90%** inference cost savings through **pipeline profiling** and optimization.

PROJECTS

SoloPilot | *Python, TypeScript/Next.js, AWS, LangChain*

Personal Project

- Prototyping an **AI-powered** freelancer workspace with **agentic workflows** automating intake, scope analysis, proposal generation, and delivery.
- Designed an autonomous **evaluator→revisor agent loop (LangChain/Python)** with rubric-based **benchmarking**, improving quality through measured **AI performance** analysis.
- Shipped an email intake console rendering versioned proposal PDFs via a **Vision Model**, demonstrating practical **AI integration** into day-to-day workflows.

Backer | *TypeScript/Next.js, AWS DynamoDB, NextAuth*

Hack The Coast 2026

- Architected an **AI-powered** startup discovery platform (**TypeScript/Next.js, DynamoDB**) serving personalized investor feeds at scale.
- Built **ML training infrastructure**: supervised pipeline ingesting signals from **DynamoDB event streams** to train an **XGBoost** ranker.

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Python, Java, SQL, C/C++, Haskell (coursework), HTML/CSS

AI & Platforms: OpenAI API, LangChain, NLP/embeddings; Next.js, Node.js, Playwright; AWS (Lambda, DynamoDB, S3, SQS, CloudWatch); Git, Docker, CI/CD